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Restructuring Local Public Transportation Through Ride-Sharing by Driving Schools in Japan: A Policy Proposal Drawing on International Contexts

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Abstract: This study proposes a policy framework to restructure local public transportation in rural Japan through a demand-responsive ride-sharing system operated by local driving schools. Faced with severe driver shortages and unprofitable bus routes, Japan requires sustainable mobility alternatives. Through a narrative review of international ride-sharing evidence and a supplementary survey of user preferences in Thailand, we identify safety, operational reliability, and affordability as critical for user acceptance. To address concerns from the domestic taxi industry regarding safety and liability, we propose driving schools as highly suitable operating entities, given their professional instructors, rigorous safety standards, and existing dispatch capabilities. Furthermore, this study presents a preliminary qualitative feasibility assessment addressing organizational bottlenecks, such as capacity conflicts during peak training seasons, and proposes practical countermeasures. Finally, it argues that local governments should redirect existing subsidies—currently used to cover traditional bus deficits—toward the driving-school demand-responsive transport (DRT) model. Ultimately, this policy proposal offers a viable, safety-conscious mechanism to maintain public transportation as an essential service for social inclusion and minimum mobility guarantees while ensuring sustainable mobility in regional areas.

Keywords: Local public transportation; Ride-sharing; Driving school; Policy proposal; Demand-responsive transport; Driver shortage; Traffic safety; Regional mobility

1 Introduction

This study examines and proposes the restructuring of local public transportation through ride-sharing in Japan. The management of shared buses has not been sustainable owing to a lack of riders, despite increased ridership and fare revenue. The driver shortage has led to reduced services in otherwise profitable urban routes.

The average age of bus drivers in Japan is higher than that of the national industry, with the female ratio below 4%. Long working hours and low income, compared with the industry average, create harsh conditions that deter younger workers, both men and women, from this position. Bus operators rarely expend the effort to attract high school graduates seeking employment or improve the working environment for all employees, especially women [1].

While the total number of industry workers has been increasing, the numbers in the motor transport business, such as bus and truck driving, have flattened. Some bus operators have, thus, been forced to reduce services due to driver shortages [1].

In early April 2024, the Work Style Reform Law introduced upper limits on overtime in order to prevent long working hours and improve working conditions, especially for bus drivers. However, the law forces a reduction in driving hours per driver, exacerbating the driver shortage. The regulation of overtime labor without a resolution of the labor shortage has led to reduced services and route cancellations, squeezing profits not only for users but also for operators and drivers and forcing bus operators to withdraw from specific areas.

Many municipalities now operate community buses, covering the operating costs in areas where the bus route has already been discontinued or become unprofitable. Sustaining this kind of operation through municipal funding alone is difficult. Ogata et al. [2], for instance, suggest the necessity of beneficiary-funded operation.

It is imperative to explore ways to provide public transportation that reduces the burden on local governments while ensuring equal access to public transportation. In response, local governments and non-profit organizations (NPOs) are taking the lead to provide public transportation through private paid passenger transportation services. In this study, we describe "private-use paid passenger transportation" and the newly launched "private-vehicle utilization service" as "ride-sharing."

The private paid passenger transportation system allows municipalities and NPOs to provide paid transportation using private vehicles. While normally prohibited, it has recently been permitted as an alternative means of transportation when essential services for maintaining residents' livelihoods, such as transportation in depopulated areas or welfare transportation, cannot be provided by bus or taxi services. It thus fulfills a role in local public transportation. We refer to these services as "municipal ride-sharing."

For example, in Fukui Prefecture, following the discontinuation of the bus route connecting Ikeda Town and neighboring Fukui City at the end of March 2019, the town launched a community-collaborative service using wagons and minibuses starting April 1, 2019, to ensure daily transportation [3]. The deficit compensation to the municipality is still higher than the operating costs of the town-resident collaborative bus. Thus, the municipality bears the majority of the operating costs.

Similarly, as an example of ride-sharing services operated by an NPO, Kyotango City in Kyoto Prefecture started the "Mutual Support Transportation Service" [4]. The city was established on May 26, 2016, through the merger of six towns: Tango, Yasaka, Omiya, Mineyama, Amino, and Kumihama. The service uses the Uber Japan system to provide transportation using private vehicles driven by local residents. This is the first instance of private car transportation using an Uber system. The fares are more affordable than those of taxis. The service is, however, limited to areas within Tango town and to travel between Tango town and other towns within Kyotango city. The Mutual Support Transportation Service operates without government subsidies; it is sustained solely by fare revenue, with the fares being around half the cost of taxi fares.

Launched in April 2024, the Private Vehicle Utilization Project addresses taxi shortages. It permits paid transportation services using local private vehicles and general drivers in areas where taxis are insufficient. The service benefits from Article 78, Item 3, of the Road Transport Act, which allows exceptions "when the taxi is unavoidable for the public welfare." This program, termed "Japanese-style ride-sharing," operates under taxi operator oversight and is limited to specific regions, periods, and times. It is primarily restricted to early mornings and evenings but unavailable during daytime hours. Fares are comparable to those of taxis.

To avoid conceptual ambiguity, it is necessary to establish operational definitions for the various transportation models discussed in this study. We classify them into the following four categories: Municipal ride-sharing, Japanese-style ride-sharing, Commercial ride-hailing (e.g., Thailand), and demand-responsive transport (DRT). Table 1 outlines their operating entities, legal bases, pricing, and regulations.

Table 1. Classification of ride-sharing and related transport concepts

Concept	Operating Entity	Legal Basis	Pricing	Primary Purpose/Regulation
Municipal ride-sharing (Private-use paid passenger transport)	Municipalities and non-profit organizations (NPOs)	Japan: Road Transport Act (Article 78, Item 2)	Subsidized (often cheaper than taxis)	Ensuring mobility in depopulated areas. Strictly regulated for welfare purposes.
Japanese-style ride-sharing (Private-vehicle utilization)	Taxi companies	Japan: Road Transport Act (Article 78, Item 3)	Same as standard taxi fares	Addressing taxi shortages in specific areas/times. Operates under taxi company oversight.
Commercial ride-hailing (e.g., Thai Grab/Bolt)	Private drivers via platform companies	Thailand: vehicle act (platform regulations)	Dynamic/market-driven	Profit-driven commercial transport. Fares and operating areas are generally unrestricted.
Demand-responsive transport (DRT)	Bus or taxi operators (commercial vehicles)	Japan: road transport act (article 4)	Fixed or zone-based fares	Flexible routing using commercial vehicles (not private cars) to supplement public transit.

While we focus on introducing a new framework for "ride-sharing" in Japan, we analyze the Thai context to

understand general user behaviors toward app-based private-vehicle transport.

The newly launched Japanese-style ride-sharing service aims to supplement taxi services; it is not intended for the daily commuting and all local residents. Daily mobility for older adults requires enhancing public transportation and municipal ride-sharing services. However, opposition from the taxi industry has been significant, particularly with regard to safety [5].

Thus, it is vital to establish legal frameworks for ride-sharing in order to secure transportation options for residents. For now, only the Japanese-style and municipal ride-sharing models are permitted for operation by a limited number of entities. In this study, we propose new strategies for ride-sharing as a form of public transportation. We draw on examples from overseas where ride-sharing has been permitted for much longer than in Japan. While some driving schools in Japan conventionally provide shuttle services solely for their students (e.g., the Hokuriku Driving School) or independently offer limited transport for local residents (e.g., the AOI Driving School), these remain isolated operations outside the formal public transit network. The primary novelty of this study extends beyond proposing the use of driving school vehicles to present a structural transport-integration mechanism. Specifically, we propose a framework to legally convert and integrate these driving school resources into the formal public transportation network—supported by public funding—to systematically replace financially unviable bus routes and fill mobility gaps in rural areas. The study addresses the following research questions:

- Can ride-sharing help restructure local public transportation in rural areas?
- What factors influence user preferences and acceptance of ride-sharing services?
- What type of operating entity can ensure the safety and reliability of ride-sharing?

2 Effect of Legalized Ride-Sharing on Taxis

2.1 Taxi Industry in Japan

The taxi industry is opposed to the legalization of Japanese-style ride-sharing, citing "safety"—owing to the potential of increased accidents and crime—as their primary concern. A secondary concern is the potential decrease in taxi revenue once ride-sharing increases.

However, Figure 1, which shows the traffic accident rate in Japan, indicates that the taxi accident rate is higher than that of other passenger vehicles, as well as freight and bus services. The accident rate per 100 million vehicle-kilometers driven (equivalent to 10,000 vehicles each traveling 10,000 kilometers) for taxis is more than 1.5 times that of other passenger cars [6]. Although these comprehensive data on vehicle-kilometer exposure dates back to 2010, the fundamental structural tendency of taxi accident risks remains highly relevant.

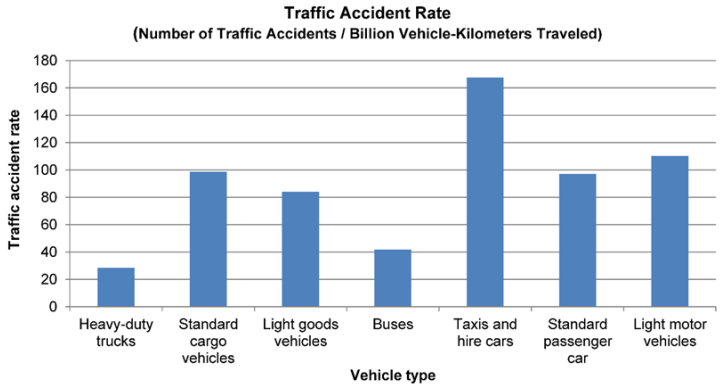


Figure 1. Traffic accident rate per vehicle-kilometer by vehicle type of primary party involved in accidents, 2010
Source: Adapted from Ref. [6] under the Creative Commons CC BY 4.0 license.

Figure 2 shows the trend in taxi accidents. Taxi accidents occur more frequently when a vehicle is empty than when it is occupied with passengers [7]. The most common causes of automobile accidents are failure to exercise due care (failure to confirm safety), failure to exercise due care (distracted driving), failure to exercise due care (failure to observe surroundings), failure to exercise due care (careless driving), and failure to proceed safely at intersections, according to the Metropolitan Police Department [8].

2.2 Accident Rates After Introduction of Ride-Sharing

In this section, rather than conducting a full systematic review, we perform a narrative review of representative international studies to evaluate the safety impacts of ride-sharing, with an aim to provide practical insights for our policy proposal.

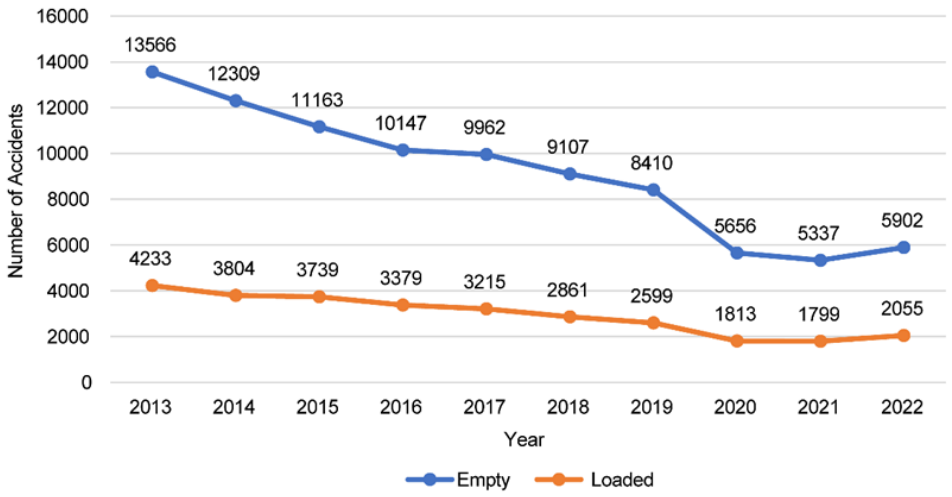


Figure 2. Trend in taxi accident counts (empty vs occupied)
Source: Traffic Accident Analysis Center (2013–2022).

In their comparison of accident rates for ride-sharing and taxi services, Zhai et al. [9] find a significant difference in serious accidents (per 1% increase in mileage: ride-sharing +0.11%, taxi +0.10%). However, ride-sharing accidents led to more minor injuries (per 1% increase in mileage: ride-sharing +0.17%, taxi +0.07%), possibly because of the need to operate the ride-sharing app while driving. While a significant increase in accidents is not a major concern for introducing ride-sharing in Japan, the handling of communication devices requires greater caution.

The introduction of ride-sharing has alleviated taxi shortages, reduced drunk driving, and decreased serious accidents across countries. In Madrid, Spain, ride-sharing led to a 25% decrease in serious injuries and fatalities across the city based on data from 2013 to 2019 [10]. However, minor injuries increased by about 5%. Kirk et al.'s [11] comparison of areas with and without ride-sharing in the UK, covering 378 areas from 2009 to 2017, shows that ride-sharing reduced serious injuries by 9% nationwide. It contributed to a 20% reduction in London. Minor injuries decreased by 7% outside London but increased by 18% in London.

Overall, the international evidence presents a clear trade-off: ride-sharing tends to decrease serious accidents (often attributed to reduced drunk driving) but increases minor collisions (likely due to driver distraction from dispatch apps, unfamiliarity with routes, or a gross increase in traffic volume). Therefore, addressing the Japanese taxi industry's concerns requires acknowledging this duality. For Japan to successfully implement ride-sharing, the increase in minor accidents observed in Madrid and London highlights the need for preventative measures, including strict restrictions on smartphone use while driving and mandatory driver training on app usage.

2.3 Effect of Ride-Sharing on Taxi Revenue

Evidence from other countries shows no major decrease in taxi revenue, which is one factor for opposition to ride-sharing by taxi drivers and operators. For instance, in New South Wales, Australia, the taxi market has not shrunk, and overall revenue by door-to-door passenger transport has increased [12]. Thus, taxis and ride-sharing can coexist. Covering the period of 2009 to 2019 in the U.S., Tashiro and Choi [13] show that ride-sharing platform Uber's introduction had no significant effect on either the labor supply or income of taxi drivers.

Thus, ride-sharing secures door-to-door transportation across a wider area, increasing the mobility of all residents and visitors. Rural Japan, where public transportation is not easily accessible, is especially in need of ride-sharing options.

However, these overseas findings should not be generalized to Japan. The Japanese taxi market structure, road infrastructure, and legal liabilities differ significantly from those in the U.S. or Europe. Although the "Japanese-style ride-sharing" project launched in April 2024 aims to integrate with the existing taxi industry, comprehensive empirical data or early policy evaluations are lacking due to its relatively recent implementation. While the lack of domestic pilot data restricts definitive conclusions, the overseas literature fundamentally demonstrates that, under appropriate regulatory frameworks, ride-sharing can coexist with existing transport networks without inherently suppressing them. Future research must evaluate domestic operational data once it matures to validate these extrapolated insights.

In the following, we present a survey case study of ride-sharing in Thailand in order to propose the prerequisite conditions for promoting ride-sharing in Japan.

3 Survey on the Ride-Sharing in Thailand

3.1 Ride-Sharing in Thailand

Ride-sharing through private vehicles is legally permitted in Thailand. These services are operated via popular apps, such as Grab and Bolt. While their use is completely unrestricted by time and geography, their fares can be higher than the fares for metered taxis. The initial fare is 35 baht for a metered taxi and 40 baht for ride-sharing cars (based on the author's field investigation in March 2024). The fare is fixed when the destination is set, and payment is cashless. Thai taxis, like those in Japan, operate on a meter system, meaning the fares are not subject to whims. However, illegalities, such as drivers demanding arbitrary prices or refusing passengers, are commonplace. A clear fare structure and a reliable destination arrival have gained support among people, leading to the rapid expansion and growth of ride-sharing alternatives.

Notably, Thai services like Grab and Bolt must be defined as "commercial ride-hailing" rather than "ride-sharing" in the strict Japanese sense. While Japanese ride-sharing is heavily regulated to supplement public transit gaps (welfare-oriented), Thai ride-hailing operates as a profit-driven commercial service with dynamic pricing and fewer operational restrictions. Despite their regulatory differences, their user experience remains similar; both utilize private vehicles, non-professional drivers, and app-based dispatch systems. By analyzing this Thai commercial model, we can extract fundamental user preferences—such as the demand for safety, reliability, and app usability—that are universally applicable and particularly necessary for designing a successful ride-sharing system in Japan.

3.2 Questionnaire and Results

We conducted a questionnaire survey in March 2024 on ride-sharing among students and citizens near Thammasat University's Rangsit Campus in order to understand the respondents ride-sharing habits and desired features in ride-sharing. The Rangsit area, where Thammasat University is located, is situated about an hour's drive north of the capital, Bangkok. It is a suburban area with large shopping malls. The area lacks railways, making cars the primary means of transportation—a situation similar to that in Japan's regional areas. Table 2 shows an overview of the ride-sharing survey in Rangsit District. The respondent attributes are as follows: of the total, 21 were men and 35 were women; two respondents selected "other" gender (Figure S1). Most respondents were in their 20s, with students constituting the majority (Figures S2 and S3).

Table 2. Overview of the ride-sharing survey in Rangsit District, Thailand

Category	Item	Description
Survey information	Survey location	Near Thammasat University Rangsit Campus, Thailand
	Survey method	Face-to-face web questionnaire survey
	Survey period	March 9–12, 2024
	Number of responses	58 people
General survey items	Attributes	–
	Car ownership	Do you own a car that you can drive?
	Primary means of transportation	–
	Experience with ridesharing	Yes/No
Branch-specific items	Yes (46 respondents)	Usage frequency Factors sought in ridesharing Preferences in ridesharing
	No (12 respondents)	Would you like to try using ridesharing? Factors sought in ridesharing Preferences in ridesharing

Note: an endash indicates "no data" or "not applicable".

Figure 3 shows the most frequently used means of transportation. Regarding private car ownership, 23 respondents (39.7%) owned a car, while 35 (60.3%), the majority, did not (Figure S4). Consequently, buses were used more frequently than cars. Cars were the second most common mode after buses. However, private cars are highly convenient and an indispensable means of transportation in suburban locations. "Rottu" and "Songthaew" are small, shared buses and trucks. Their lower usage, compared with taxis, may be due to limited boarding points and destinations.

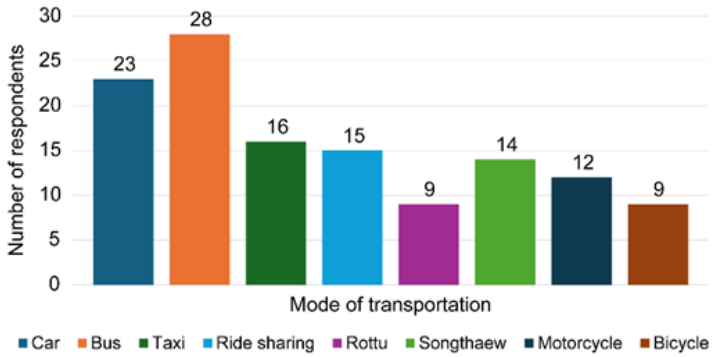


Figure 3. Most frequently used means of transportation (N = 58, multiple responses)

Figure 4 shows the frequency of ride-sharing use. Of the respondents, 46 (79.3%), the majority, used ride-sharing, while 12 (20.7%) had never used such services (Figure S5). Nevertheless, majority of the respondents only used ride-sharing "a few times a year" or "once or twice a month," indicating a very low frequency of use. For the subsequent analysis on desired features, we focused on the responses of these 46 experienced users.

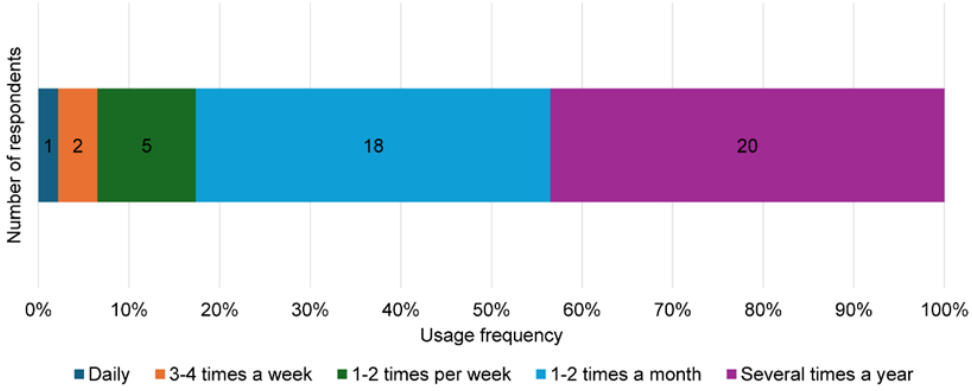


Figure 4. Frequency of ride-sharing use (N = 46)

Figure 5 shows the features of ride-sharing that respondents rated as important. The descriptive results indicate that users place high importance on safety, reliability, and price affordability, followed closely by intuitive app usability and quick pickup times. These descriptive findings suggest that when users choose a ride-sharing service, they prioritize the trustworthiness (safety and reliability) of the operating entity and the convenience and cost-effectiveness of the service.

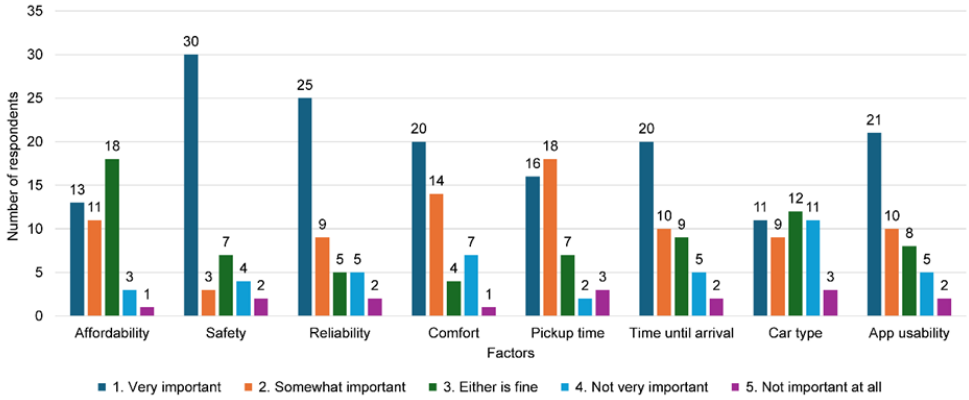


Figure 5. Desired elements in ride-sharing (N = 46)

Therefore, to successfully introduce ride-sharing as local public transportation in Japan, selecting an appropriate

operating entity that can guarantee safety and reliability is crucial. Furthermore, while the Thai survey highlights the importance of app usability, relying solely on smartphone apps may create barriers for elderly residents in rural Japan. Thus, it is vital to develop inclusive reservation methods (e.g., telephone bookings or community-assisted reservations) alongside accessible pricing supported by public funding.

In the following section, we extend the essential requirements of safety, reliability, and accessibility to propose driving schools as a novel and highly suitable operational entity for restructuring local public transportation in Japan.

While the survey sample consists mainly of Thai university students—who demographically differ from elderly residents in regional Japan—this group was surveyed to extract fundamental needs from experienced ride-share users. Since even young, accustomed users heavily prioritize safety and operator trust, these factors are logically presumed to be highly critical for elderly Japanese residents. This strongly justifies our proposal to utilize highly trusted local driving schools. Nonetheless, the demographic gap remains a limitation in external validity, warranting future studies directly targeting Japanese regional residents.

4 Discussion: New Operating Entity for Ride-Sharing

4.1 Proposal for a New Operating Entity

Under current regulations, ride-sharing operators are limited to taxi companies for Japanese-style ride-sharing and to the local governments or the NPOs for municipality-operated ride-sharing. Many concerns regarding expanding or relaxing ride-sharing regulations relate to safety issues. However, current operators cannot be definitively stated as absolute or suitable if we consider the Thai case and the current taxi accident rates in Japan. Table 3 shows a proposal for ride-sharing regulatory reform. The Mutual Support Transportation Service in Kyotango City, Kyoto Prefecture, proposes a future regulatory reform measure [14].

Table 3. Proposal for ride-sharing regulatory reform

Item	Current Paid Transportation in Public Transportation Blank Areas*	Regulatory Reform Proposal**
Organization for consensus building	Regional Public Transportation Council/Operating Council (Road Transport Act Article 79-4 Paragraph 1 Item 5, Regulation Article 51-7)	National Strategic Special Zone Council
Implementing entity	Non-profit organizations (NPOs) corporations, etc. (Article 78, Item 2 of the Act; Article 48)	Corporation
Driver	Persons listed in the driver register of NPO corporations, etc. (Article 51-19, Paragraph 1 of the Regulations)	Individuals registered with the relevant corporation
Limited-service area	Transportation-deprived areas (Article 78, Item 2 of the Act; Article 49, Paragraph 1, Item 2 of the Regulations)	From the user's perspective, regional councils will certify cases where public transportation does not provide sufficient service; service implementation will be enabled in the designated area. Example: This includes suburban new towns, major arterial roads during commuting hours, and addressing tourism or special demand needs, in addition to existing transportation gaps
Confirmation, recording, and roster management for safe driving	In principle, conducted in person or in writing (Ministry of Land, Infrastructure, Transport and Tourism processing policy, etc.)	Permits confirmation, recording, and roster management for safe driving using information and communications technologies

Note: * The current regulations refer to the Public Transportation Act (Article 78, Item 2; Article 79-2, Paragraph 1, Item 2; Article 49, Paragraph 1, Item 2; Article 51, Item 2, etc.). Act: Road Transport Act; Regulations: Road Transport; ** The proposed regulatory reforms aim to modify the existing paid transportation framework in areas not adequately covered by current services by establishing a new category, "Fourth Category of Paid Passenger Transportation by Private Vehicles," with the regulatory provisions outlined in the table. Source: Adapted from Ref. [14].

This proposal revises the current regulations governing paid transportation in areas without public transit. The limitations on operating entities and service areas are notable for being significant constraints for both municipal and Japanese-style ride-sharing. Future operators should be selected based on their appropriateness for local

circumstances; this is crucial, as evidenced by the Thai survey. The Thai findings highlight the paramount importance of a trusted operating entity for perceived safety. Accordingly, we hypothesize that organizations with an established reputation for traffic safety could fulfill this role in Japan. Although this study does not include direct survey data on Japanese residents' trust in specific entities, we logically propose driving schools as a highly suitable candidate based on their institutional characteristics.

Most driving schools are designated as driving instruction facilities. They are managed by prefectural public safety commissions, which are part of the police, based on Article 99 of Road Traffic Act. These schools possess a high level of awareness of traffic safety. Kaga City in Ishikawa Prefecture, which operates a municipal ride-sharing service through Uber Japan's platform, also collaborates with driving schools [15].

In March 2024, Kaga City signed an agreement of comprehensive partnership with the Uber Japan Inc. and launched its "Kaga City-style Ride-Sharing Service." This service is implemented through the following three-party cooperative system: Kaga City Tourism Exchange Organization serves an operating entity, Uber provides its app platform, and a taxi company Kaga Daiichi Kotsu handles operational management. This is the first ride-share service that uses privately owned vehicles for a paid passenger transport with the involvement of Uber and the local government since regulatory easing in December 2023.

Municipal ride-sharing services using privately owned vehicles as paid passenger transport differ from the private vehicle utilization projects that have been introduced since April 2024, primarily in major metropolitan areas, based on Article 78, Item 3, of Road Transport Act. These services, operated by municipalities or NPOs, provide transportation options for residents and tourists in areas with limited public transport. A fare cap for the municipal ride-sharing services was previously set at half the taxi fare. It was raised to 80% of the taxi fare following the December 2023 deregulation. Later, in October 2024, Uber Japan collaborated with Kaga Driving School for a first-ever "Safety Training Session for a Ride-Share Drivers" at Kaga Driving School. Kaga Daiichi Kotsu instructed attendees on driving safety and etiquette.

While providing safety training in Kaga City does not equate to full operational management—such as dispatching, fare collection, and regulatory compliance—it demonstrates that driving schools possess the foundational expertise and institutional trust required to support driver safety in regional ride-sharing.

Thus, driving schools are organizations with high safety awareness and are considered indispensable. Since driving schools in rural areas often operate shuttle buses to transport students, we propose expanding their use to shared transportation.

4.2 Public Transportation by Driving Schools

Shuttle bus services for local residents by driving schools have already commenced. The AOI Driving School, Tsuruga Branch, operates this service in Tsuruga City, Fukui Prefecture. This branch runs school buses on fixed routes at scheduled times. One route connects the school to central Tsuruga. The service is provided free of charge to registered local residents on a route connecting the school to central Tsuruga City. Kamimura et al.'s [16] survey study of registered and non-registered local residents covered by the Tsuruga Branch finds no correlation between perceived value and willingness to pay the same fare as a community bus. Thus, transportation services provided by driving schools are not considered conventional paid services. Shuttle services for local residents by driving schools may ideally remain a part of their corporate social responsibility activities.

4.3 Shuttle Buses as Transport Mobilization

A "Basic Policy on Revitalizing and Regenerating of Regional Public Transportation" was published in August 2023, according to the Ministry of Land, Infrastructure, Transport and Tourism [17]. The report advocates for the revitalization and regeneration of regional public transportation. This involves the sustainable provision of regional passenger transportation services to all local governments. Thus, all regional transportation resources, such as school buses, welfare transportation, and shuttle services for commercial facilities, can be used when public transportation alone cannot fully meet mobility needs. It also involves new mobility demands within a region.

Figure 6 illustrates the concept of a regional passenger transport service. This policy categorizes all local transportation services as "regional passenger transport services." That is, it aims to utilize not only conventional public transport but also local transport resources (such as privately paid passenger transport, shuttle services by schools, welfare facilities, hospitals, and commercial facilities) as part of regional public transportation. There is also a proposal to consolidate route buses while adapting them to facility needs, rather than using facility shuttle services as public transportation, according to the Ministry of Land, Infrastructure, Transport and Tourism. In any case, it is necessary to achieve effective utilization of local resources and convenience of maintenance operations. However, there remain negative opinions regarding the total mobilization of transportation resources.

A critical policy commentary by He [18] states that a comprehensive mobilization of regional transportation resources means abandoning the local government's role. This contrasts with the Ministry of Land, Infrastructure, Transport, and Tourism's perspective on comprehensive mobilization, as follows:

• The mobilization of all transportation resources generally refers to local transportation for regular passengers to use empty seats on existing shuttle buses. Local governments typically "procure public service resources through subsidies and commissions." An idea of mobilizing all transportation resources is, on the other hand, based on the premise of "getting them provided for free."

• The total mobilization of transportation resources asserts that "an area not covered by public transportation may be entrusted to the volunteer spirit of society." It is not based on the idea that "the local governments must secure necessary public transportation as with typical public services."

• The role of local governments is to solve problems that cannot be solved by the market or individuals. That is, expecting residents to ride along shuttle buses operated by facilities is not a fundamental role of local governments. It negates, furthermore, the purpose of the government.

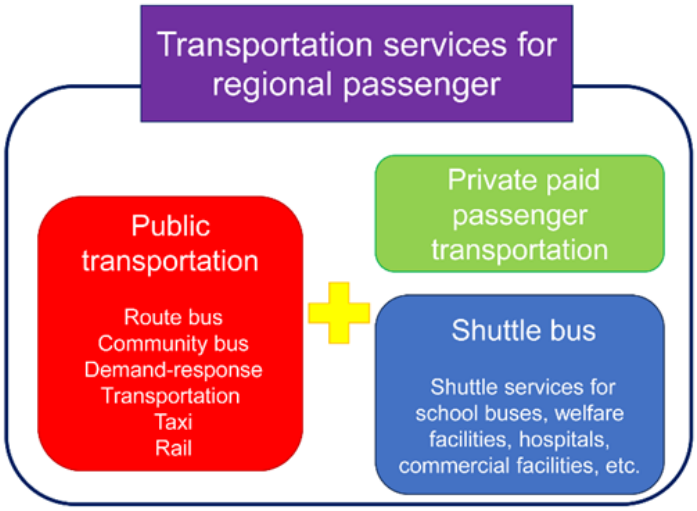


Figure 6. Regional passenger transport service concept

He [18] presents the above view and states that the fundamental role of local governments is to secure services essential to society, even if they alter resource constraints. Table 4 presents the differences between shuttle buses and public transportation. He [18] also addresses the Ministry of Land, Infrastructure, Transport, and Tourism's proposal to integrate route buses with facility shuttle buses.

Table 4. Differences between shuttle buses and public transportation

Allowing General Passengers to Ride on Shuttle Buses	Sharing Shuttle Passengers with Regular Bus Passengers
× Shuttle buses lack motivation	◯ Can comprehensively solve various societal mobility issues
× Low safety, unclear transportation liability	◯ High safety, clear transportation liability
× Shuttle buses can be discontinued at any time	◯ Municipal governance is effective

Source: Adapted from Ref. [18].

Comparing policies that allow general passengers on shuttle buses versus those that allow shuttle passengers along public transit routes on public buses, He [18] states that the latter policy is more appropriate for municipalities to adopt.

Private shuttle buses attract customers. An operator has no benefit in allowing passengers other than business customers on these buses. Attempting to include local residents in shuttle buses would rely on the operator's goodwill or require municipal funding. Furthermore, the liability for transportation is unclear, and safety cannot be guaranteed. A national government appears to view empty seats on shuttle buses as a "resource" that can be used free of charge through negotiation. Shuttle buses operate as part of a rigorous business model and are not as readily available as expected. In contrast, route buses are managed by local governments, their safety is assured, and they can comprehensively resolve mobility issues.

He [18] argues that shuttle buses, as a total mobilization of transport resources, rely on the goodwill of operators and constitute a "free rider," which represents an abdication of local government responsibility. The author proposes a model in which driving schools provide resident shuttle services. This positions shuttle buses as part of the public

transportation system. Shuttle buses operate on a regular schedule and on a demand basis, and costs are incurred for each trip. Demand-responsive services operate on a per-request basis. Therefore, the concept of providing surplus resources free of charge does not apply. Municipal interventions are, thus, essential to public transportation.

4.4 Driving Schools as Public Transportation

The Hokuriku Driving School in Fukui City, Fukui Prefecture, has operated all shuttle bus routes on a demand basis for several years in order to reduce costs. They use the VISH Inc.'s "Bus Catch" dispatch system, allowing students to pinpoint reserve pickup locations via an app according to their daily schedule. Figure 7 shows the pick-up and drop-off areas on the Bus Catch screen. According to the Hokuriku Driving School website, pickup areas cover a wide region in the northern part of Fukui Prefecture, extending 20 km from the school [19]. Students can reserve pickups at various destinations from the Hokuriku Driving School via the app.

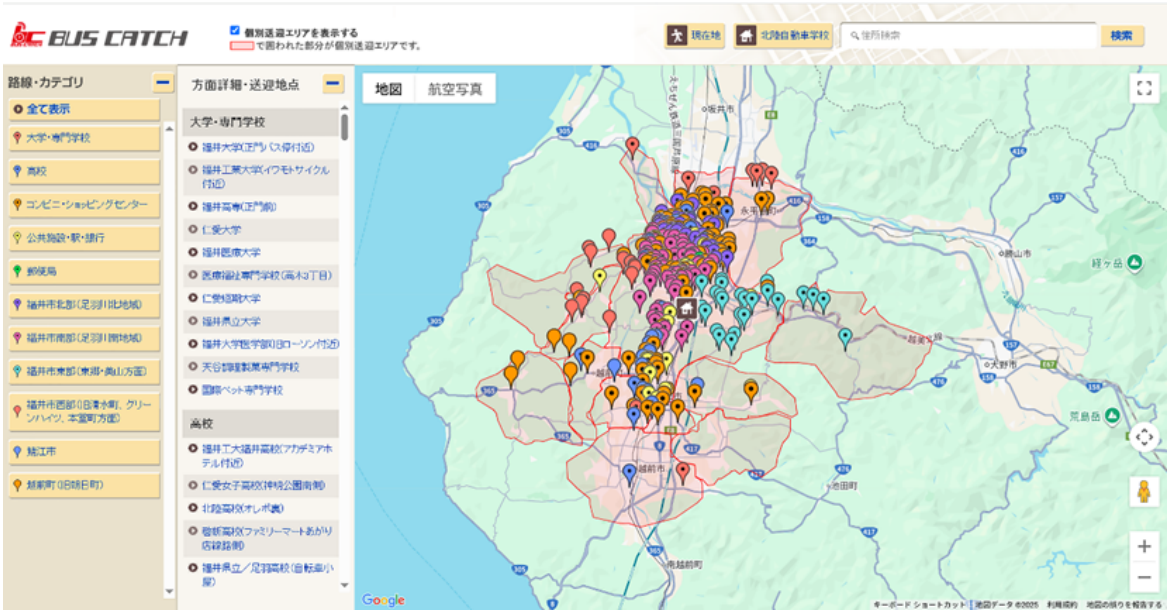


Figure 7. Pickup/drop-off area via Bus Catch screen
Source: Reproduced from Ref. [19].

The benefits for students include the freedom to choose pick-up or drop-off locations near their homes, schools, and part-time jobs. It enables door-to-door commuting, like a taxi, because there is no fixed route. Demand-responsive services allow reservations at the desired times.

The benefits for schools and drivers include eliminating empty runs, enabling optimal route planning, and reducing fuel waste and labor costs. There is no risk of overlooking passengers waiting along the route, because the number of passengers per trip is known in advance. The author conducted follow-up interviews on August 28, 2026, with the president of Hokuriku Driving School, who reported that converting the shuttle bus to a demand-based service resulted in annual savings of approximately 8 million yen. While an exact breakdown was unavailable, this saving is equivalent to the cost of 2.8 employees when labor and fuel costs are considered. A shift to demand-based operations is considered highly effective.

Let us clarify the differences between fixed-and demand-based shuttle bus operations, as follows:

Scheduled Routes. Each bus has a fixed schedule and route, and operates regardless of the presence or number of passengers. Local residents can use the service at no additional cost, as long as seats are available.

Demand-Based Service. These operations are based on student reservations; therefore, routes and destinations constantly change. There may be no trips available depending on the time or destination. There is no surcharge when local residents use the service and the time/destination matches. However, a fee is charged each time only local residents use it.

The cost remains unchanged when the shuttle buses operate along fixed routes. Driving schools that provide transportation for local residents can then be considered part of their corporate social responsibility activities, as suggested by Kamimura et al. [16]. However, costs arise in the case of demand-responsive operations if students cannot be picked up together. This creates a free-riding situation that diminishes the effectiveness of demand-responsive services initiated to reduce costs. Therefore, it is necessary to operate them as business ventures when local residents use demand-responsive shuttle buses.

To resolve the tension between leveraging idle resources and maintaining service quality, we address the critiques raised by He [18] regarding the "free-rider" exploitation of surplus transport capacity. He outlines the following three major concerns: ambiguous operational motivation, unclear safety liabilities, and high risks of service discontinuation. The fee-based DRT model operated by driving schools systematically overcomes these vulnerabilities. First, regarding operational motivation, the fee-based structure shifts the service from a purely altruistic corporate social responsibility initiative to a sustainable business diversification strategy. Charging a fee explicitly covers variable costs (e.g., fuel and vehicle wear), transforming idle resource utilization into an economically rational operation. Second, concerning safety and liability, driving schools possess inherent institutional advantages over informal free-rider models. Driving schools are highly regulated entities equipped with professionally trained instructors acting as drivers, rigorously maintained fleets, and comprehensive commercial liability insurance. This professional framework is proposed to significantly mitigate the ambiguity of accident liability. Finally, regarding sustainability, transitioning from a free to a fee-based model could support long-term viability. By structuring operations as a potentially financially self-sustaining model tied directly to user demand, the service could be better insulated from the risk of sudden discontinuation that typically plagues models relying solely on temporary idle capacity or surplus subsidies. These propositions—economic sustainability, liability management, and cost advantages—require testing in future pilot implementations.

To address the inefficiencies of fixed routes, municipalities are exploring alternatives, such as DRT (using commercial vehicles) or municipal ride-sharing (using private vehicles). However, merely changing the model may not be sufficient. For instance, passenger numbers remained stagnant despite the introduction of community buses following the reduction in route bus services and the provision of free fares in Minamiosumi Town, Kagoshima Prefecture [20]. The town has an aging rate of 51.3%, with those aged 65 and over constituting the majority. Elderly residents prefer to rely on taxis or friends' cars, even if they cost money, because it involves less walking. The priority for older adults is not free or low-cost fares but door-to-door transportation. The chairman of a neighborhood association in the Odomari district of the town also stated, "What is desired is the enhancement of transportation that can take them directly from their home to their destination."

Suzuki and Ooi [21] conducted a questionnaire survey of 600 households in an urban area of Eniwa City, Hokkaido, where the "Eniwa Community Bus" operated in February 2008. The survey clarifies attitudes toward community bus usage, focusing on the perspective of older adults, by dividing age groups into three tiers. Figure 8 shows the experience of community bus use.

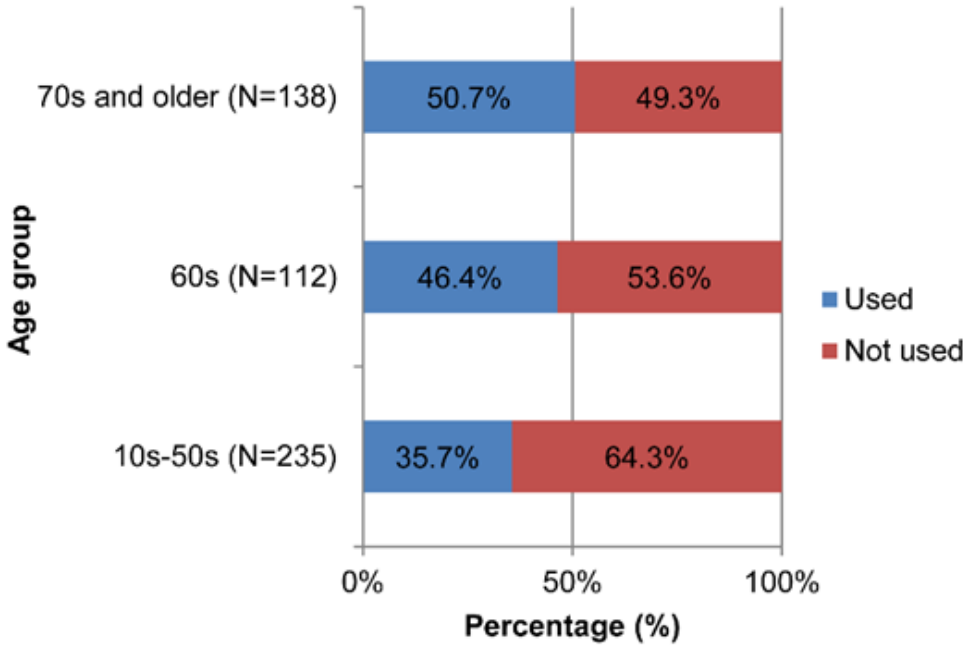


Figure 8. Experience with community bus use

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Evidently, usage increased with age. Suzuki and Ooi [21] suggest that usage is high among older adults, which indicates the need to closely monitor the needs of this demographic. However, nearly half of those aged 70 years and older did not use the community bus. The reasons for non-use were also investigated (see Table 5). Across all age groups, "Other transportation options are more convenient" was the most common response. The top three reasons

were identical for participants in their 10–50s and 60s, indicating similar attitudes. However, attitudes differed between those in their 60s and 70s because of physical concerns.

Table 5. Reasons for not using community buses

Rank	10s–50s	60s	70s
1st	Other transportation options are more convenient (34.9%)	Other transportation methods are more convenient (33.3%)	Other transportation methods are more convenient (33.3%)
2nd	Few departures (12.9%)	Few flights (13.7%)	Doesn't go where I want to go (12.0%)
3rd	Doesn't go where I want to go (10.8%)	Doesn't go where I want to go (13.7%)	Routes/schedules are hard to understand (11.0%)
4th	Routes and schedules are difficult to understand (8.7%)	Long travel time (8.8%)	Bus stops offer no shelter from wind/rain (10%)
5th	Poor transfer connections (6.2%)	Routes/schedules are difficult to understand (7.8%)	Too few departures (6.0%)

Source: Adapted with permission from Ref. [21] © 2009 Publisher.

Community buses in rural areas operate under the assumption that they are used by older adults who cannot drive themselves. However, in both Minamiosumi Town in Kagoshima Prefecture and Eniwa City in Hokkaido, Suzuki and Ooi [21] show that, even when community bus routes exist within their living areas, older adults may still find it difficult to use them. Thus, older adults prioritize door-to-door transportation. It may then be necessary to develop services that consider age group as a demographic. Building demand-responsive or ride-sharing transportation systems could potentially make modes of transport easier for older adults to use.

The reasons for proposing driving schools as operators of public transportation services are as follows. Driving schools are managed by police organizations and are highly conscious of traffic safety. They already have systems that are capable of providing wide-area transportation. There is potential for earlier commercialization compared with starting a business from scratch. Moreover, driving schools have established systems to efficiently transport people within a 20 km radius within one to two hours.

To clarify the practical integration of this service, the proposed driving-school DRT is intended to primarily substitute for low-demand, financially unviable bus routes in rural areas. It may act as a feeder service connecting to major bus or rail transfer points where trunk routes remain viable. Basic coordination with the remaining regional transport network would involve aligning DRT operating periods with existing bus or train schedules to facilitate smooth transfers at designated transit hubs. Fares would be set at a flat or zone-based rate comparable to existing community buses to ensure affordability. Crucially, to ensure accessibility for the elderly, reservation channels would include traditional telephone bookings and community-assisted reservations at local facilities (e.g., clinics or community centers), rather than smartphone applications alone. This integration model aims to complement rather than disrupt the existing essential transit network.

4.5 Issues in Commercialization

The legal issue for driving schools to operate ride-sharing as a business is that it is now only permitted for municipalities, NPOs, and taxi companies. The regulations governing ride-sharing operations are under review, with proposals for revision in the Diet. Implementation is only possible through regulatory amendments. However, there are operational challenges beyond these regulations.

Organizationally, a significant operational risk is the capacity conflict during peak training seasons (e.g., February and March in Japan). As the availability of shuttle vehicles becomes a major bottleneck during these periods, operational adjustments are required to prevent the public transit service from crowding out the school's core business. Feasible countermeasures include limiting the DRT slots during peak training hours or temporarily outsourcing overflow demand to local taxi operators through a supplementary agreement.

Furthermore, while detailed cost-benefit models and precise demand forecasting are essential for full-scale implementation, they require localized pilot data that is currently unavailable. Future pilot programs must comprehensively assess technical, economic, legal, and organizational dimensions before nationwide rollout.

Public transportation in rural areas operates on the premise of loss. The costs for local residents could exceed fare revenue if driving schools operate independently. Providing this service to residents can burden the core operations of a driving school and potentially strain its finances. Thus, public transportation requires administrative support to

maintain regional traffic safety and ensure mobility. Securing mobility for vulnerable groups, such as students, is a critical administrative responsibility. For example, recognizing this public welfare necessity, Fukui City introduced a free-trial bus service specifically for high-school students commuting to school after the discontinuation of the Keifuku Bus Gumizaki Line in October 2024 [22].

As Yukifumi Yoshikawa, President of Fukui Railway, stated, "All of our route bus lines are operating at a loss. Administrative support covered these deficits without generating profits. We ask for your understanding and support" [23]. Therefore, it is extremely difficult to maintain public transportation in rural areas. The following shows Fukui Railway's (April 1, 2023–March 31, 2024) financial situation from its most recent securities report.

- Securities report—109th Fiscal Year (April 1, 2023–March 31, 2024)
- Consolidated management indicators net loss attributable to parent company shareholders: 56.815 million yen.
- Submission of company management indicators net loss for the period: 35.713 million yen.

Fukui Railway operates at a net loss of 35.713 million yen, with consolidated deficits reaching 56.815 million yen. The government currently bears a significant proportion of these deficits through subsidies to maintain operations. Rather than simply continuing to subsidize increasingly underutilized traditional bus routes, the scale of these existing subsidies can serve as a financial reference point for funding new operational models. By redirecting a portion of these current public transit subsidies to driving schools, municipalities could establish a more cost-effective, demand-based ride-sharing network. If the required subsidy for a driving-school DRT system is lower than the current deficit of traditional route buses, this restructuring would simultaneously ensure regional mobility and optimize the financial burden on local governments.

5 Conclusions

In this study, we discuss new strategies for utilizing ride-sharing based on examples from foreign countries where ride-sharing was legalized earlier than in Japan. We also review the current state of public transportation in Japan.

Our research questions are addressed as follows: (1) ride-sharing can fundamentally restructure local public transport by filling mobility gaps where traditional buses operate at a loss; (2) based on our findings, user acceptance and preferences are primarily driven by the trustworthiness of the brand, safety, and price affordability; and (3) finally, driving schools, possessing strong institutional safety awareness and dispatch capabilities, emerge as highly suitable operating entities to ensure regional transport reliability. Ultimately, our proposal aims to establish a new governance and integration model, rather than merely extending existing private shuttle services. By formally repositioning driving schools as legally recognized public transport operators under a publicly subsidized demand-responsive framework, this study offers a viable, safety-conscious mechanism to rebuild sustainable regional mobility.

Ride-sharing originated with Uber Technologies, founded in Silicon Valley, USA, in 2009, and has spread globally across the U.S., as well as in Europe, China, Southeast Asia, and beyond. Ride-sharing became widespread before legal frameworks were established; therefore, in major countries, legal frameworks were developed in parallel with the adoption of ride-sharing. In Japan, ride-sharing as a private vehicle utilization service has been permitted since 2024, and the country is currently formulating regulations in parallel.

Studies on ride-sharing outside Japan suggest that the introduction of ride-sharing may contribute to reductions in serious crashes, while effects on minor collisions are mixed and context-dependent. However, it has not been shown to decrease taxi revenue. This case study of Thailand, in particular, reveals that users place considerable importance on reliable operators and convenient ride-sharing services. Thus, it is important to view ride-sharing as a means of transportation that sustains local communities within future legal frameworks. A key challenge may be to utilize locally supported ride-sharing as part of public transportation.

Furthermore, securing mobility for all residents, particularly vulnerable groups such as the elderly and students, is vital for social inclusion. Transportation service provision by administrative bodies is essential when it meets residents' needs. Depriving students or elderly residents of means of commuting owing to the loss of previously existing transportation is unacceptable from a public welfare perspective. Accordingly, ensuring the availability of transportation options promotes social equity and can prevent population outflow (the lack of public transportation in rural areas may lead to an outflow of young people seeking convenience, resulting in rural depopulation).

Maintenance of public transportation is currently facing nationwide difficulties that have varied sources. Public funding may be essential for this upkeep. Such funding should be meaningfully utilized. Investing public funds in community buses that fail to meet the needs of older adults, for example, is an inefficient spend. We propose driving schools as a potentially viable, safety-conscious ride-sharing alternative to current public transportation. However, it may be necessary to consider reconstructing public transportation to fit local realities. These findings may be interpreted as indicative rather than definitive, paving the way for more inclusive local transit policies.

Author Contributions

Conceptualization, K.O.; methodology, K.O. and T.T.; validation, K.O.; formal analysis, K.O.; investigation, T.T.; resources, S.Y. and T.T.; writing—original draft preparation, K.O.; writing—review and editing, K.O., S.Y.,

and T.T.; supervision, K.O.; project administration, K.O.; funding acquisition, K.O. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI was used solely to assist in the visual design and layout of the graphical abstract. All scientific content, interpretation, and final verification were performed and approved by the authors.

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Appendix

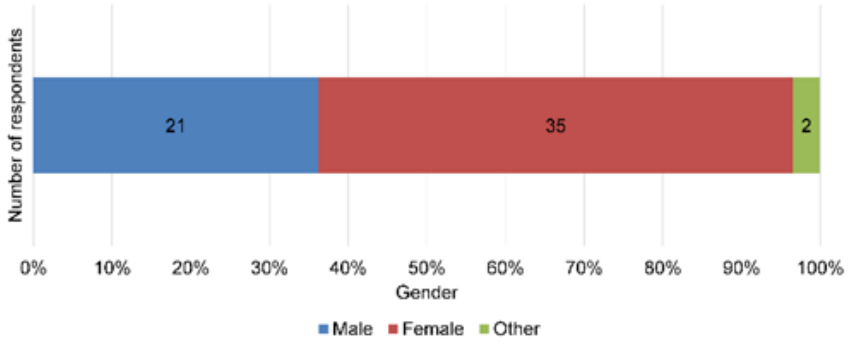


Figure S1. Gender (N = 58)

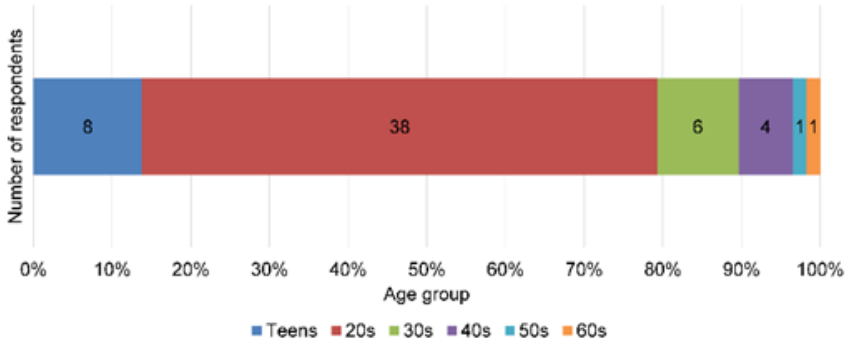


Figure S2. Age (N = 58)

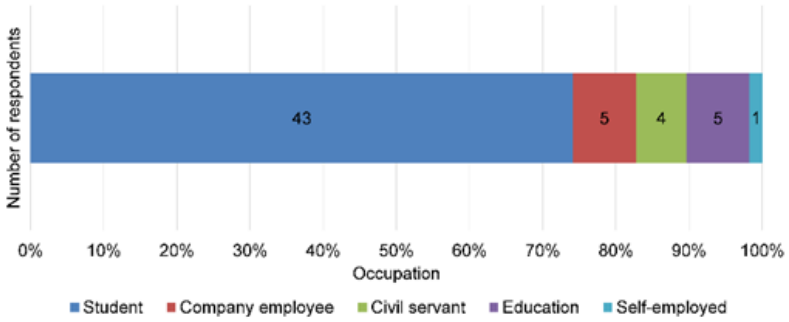


Figure S3. Occupation (N = 58)

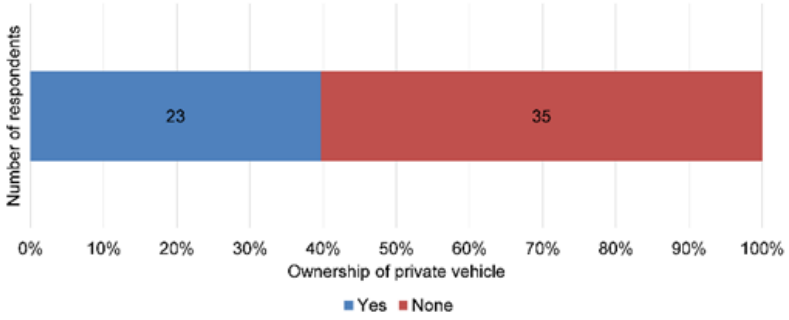


Figure S4. Ownership of private vehicle (N = 58)

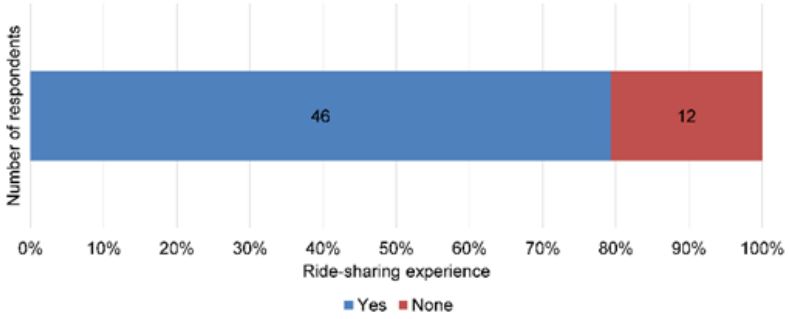


Figure S5. Experience with ride-sharing (N = 58)